

CONCEPT 21.6

Comparing genome sequences provides clues to evolution and development (pp. 459–464)

- Comparisons of genomes from widely divergent and closely related species provide valuable information about ancient and more recent evolutionary history, respectively. Analysis of single nucleotide polymorphisms (SNPs) and copy-number variants (CNVs) among individuals in a species can also shed light on the evolution of that species.
- Evolutionary developmental (**evo-devo**) biologists have shown that homeotic genes and some other genes associated with animal development contain a **homeobox** region whose sequence is highly conserved among diverse species.

? What type of information can be obtained by comparing the genomes of closely related species? Of very distantly related species?

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

1. Bioinformatics includes
 - (A) using DNA technology to clone genes.
 - (B) using computer programs to align DNA sequences.
 - (C) using a person's genomic sequence to inform decisions about medical treatment.
 - (D) amplifying DNA segments from a species' genome.
2. Homeotic genes
 - (A) encode transcription factors that control the expression of genes responsible for specific anatomical structures.
 - (B) are found only in *Drosophila* and other arthropods.
 - (C) are the only genes that contain the homeobox domain.
 - (D) encode proteins that form anatomical structures in the fly.

Levels 3-4: Applying/Analyzing

3. Two eukaryotic proteins have one domain in common but are otherwise very different. Which of the following processes is most likely to have contributed to this similarity?
 - (A) gene duplication
 - (B) alternative splicing
 - (C) exon shuffling
 - (D) random point mutations
4. **DRAW IT** Below are the amino acid sequences (using single letters; see Figure 5.14) of three short segments of the FOXP2 protein from five species. These segments contain all amino acid differences between the FOXP2 proteins of these species. Compare the amino acid sequences by answering parts (a)–(d).

Chimpanzee	PKSSD ... TSSTT ... NARRD
Mouse	PKSSE ... TSSTT ... NARRD
Gorilla	PKSSD ... TSSTT ... NARRD
Human	PKSSD ... TSSNT ... SARRD
Rhesus monkey	PKSSD ... TSSTT ... NARRD

- (a) Circle the names of any species that have identical amino acid sequences for the FOXP2 protein.
- (b) In the sequence for the mouse, circle any amino acid that differs from the sequence for the chimpanzee, gorilla, and rhesus monkey. Then draw a box around any amino acid that differs from the human sequence.
- (c) In the human sequence, underline any amino acid that differs from the sequence for the chimpanzee, gorilla, and rhesus monkey.

- (d) Primates and rodents diverged about 65 million years ago, and chimpanzees and humans diverged about 6 million years ago (see Figure 21.17). How many amino acid differences are there between the sequence for the mouse and the sequence for the chimpanzee, gorilla, and rhesus monkey? How many amino acid differences are there between the human sequence and the sequence for the chimpanzee, gorilla, and rhesus monkey? Based solely on the numbers of amino acid differences occurring over these time periods, what might you hypothesize about the rate of evolution of the FOXP2 gene? Based on the information in the chapter regarding the FOXP2 gene, is your hypothesis correct?

Levels 5-6: Evaluating/Creating

5. **EVOLUTION CONNECTION** Genes important in the embryonic development of animals, such as homeobox-containing genes, have been relatively well conserved during evolution; that is, they are more similar among different species than are many other genes. Explain why this is.
6. **SCIENTIFIC INQUIRY** The scientists mapping the SNPs in the human genome noticed that groups of SNPs tended to be inherited together, in blocks known as haplotypes, ranging in length from 5,000 to 200,000 base pairs. There are as few as four or five commonly occurring combinations of SNPs per haplotype. Integrating what you've learned throughout this chapter and this unit, propose an explanation for this observation.
7. **WRITE ABOUT A THEME: INFORMATION** The continuity of life is based on heritable information in the form of DNA. In a short essay (100–150 words), explain how mutations in protein-coding genes and regulatory DNA contribute to evolution.
8. **SYNTHESIZE YOUR KNOWLEDGE**



Insects have three thoracic (trunk) segments. While researchers have found insect fossils with wings on all segments, modern insects have wings or related structures on only the second and third segment. In modern insects, *Hox* gene products act to inhibit wing formation on the first segment. The treehopper insect (above) is an exception. In addition to the pair of prominent wings on its second segment, its first segment has an ornate helmet resembling a set of thorns, which is a modified, fused pair of “wings.” (This provides camouflage in tree branches, reducing the risk of predation.) Explain how changes in gene regulation could have led to the evolution of such a structure.

For selected answers, see Appendix A.

Explore Scientific Papers with Science in the Classroom 

How do transposable elements drive evolution?

Go to “Jumping Genes!” at www.scienceintheclassroom.org.

➔ **Instructors:** Questions can be assigned in Mastering Biology.